Cisco Catalyst 9115AXI Access Point Manual

1. Official Cisco Sources

This manual is based on official Cisco documentation for the Cisco Catalyst 9115 Series Wi-Fi 6 Access Points, especially the Cisco Catalyst 9115 Series Wi-Fi 6 Access Points Data Sheet, the Cisco Catalyst 9115AX Series Access Point Getting Started Guide, and the Catalyst 9115AX Series support and documentation portal.

The uploaded source manual identifies the Cisco Catalyst 9115AXI as an indoor enterprise Wi-Fi 6 access point with internal antennas, 4x4:4 MIMO, Bluetooth Low Energy 5.0, a multigigabit Ethernet uplink, PoE/PoE+ power support, RJ-45 console access, USB 2.0, and controller-based management.

Practical deployment, setup, verification, and troubleshooting notes were added to make this manual more useful for installation and support work. Exact commands and available features can vary depending on controller type, software release, AP mode, and licensing, so they should always be verified against the actual Cisco controller and access point software in use.

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2. Device Description

The Cisco Catalyst 9115AXI is an indoor enterprise Wi-Fi 6 access point from the Cisco Catalyst 9100 family. It supports IEEE 802.11ax wireless networking and is designed for high-density indoor environments such as offices, meeting rooms, classrooms, hospitals, hotels, and smart buildings.

The “AXI” model uses internal antennas, which makes it suitable for clean indoor ceiling or wall-mounted installations where external antennas are not required.

The access point provides wireless connectivity for client devices such as:

- Laptops
- Smartphones
- Tablets
- IP phones over Wi-Fi
- Wireless barcode scanners
- Meeting room devices
- Smart office devices
- IoT devices
- Guest users
- Corporate users

The access point connects to the wired network through its multigigabit Ethernet port. This uplink normally connects to a PoE-capable switch, such as a Cisco Catalyst access switch. The same Ethernet cable carries both data and electrical power when PoE is used.

The Catalyst 9115AXI is usually managed by a Cisco Wireless LAN Controller, such as a Cisco Catalyst 9800 Series controller. Some EWC models can operate with an Embedded Wireless Controller.

In a smart office topology, this device should be treated as the main wireless access device. It provides Wi-Fi coverage and connects wireless clients to the wired LAN through the access switch.

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3. Technical Specifications

The Cisco Catalyst 9115AXI is manufactured by Cisco and belongs to the Cisco Catalyst 9115AX Series.

It is an indoor Wi-Fi 6 access point with internal antennas. Its main function is to provide secure wireless client access on the 2.4 GHz and 5 GHz bands.

The access point supports:

- Wi-Fi 6 / IEEE 802.11ax
- 802.11ac/n/a/b/g compatibility
- 2.4 GHz and 5 GHz wireless bands
- 4x4:4 MIMO under full power
- Up to 5.38 Gbps PHY rate under supported conditions
- Internal antennas
- Bluetooth Low Energy 5.0
- USB 2.0 port
- RJ-45 management console
- Multigigabit Ethernet uplink
- 100 Mbps, 1 Gbps, and 2.5 Gbps Ethernet speeds depending on cabling, switch support, and power mode
- Controller-based management
- Embedded Wireless Controller support on EWC models

Power requirements:

- Full functionality requires 802.3at PoE+
- Maximum power draw is approximately 20.4 W
- 802.3af PoE can power the AP but with reduced functionality
- Supported Cisco power injectors include models such as AIR-PWRINJ6= and AIR-PWRINJ5=, depending on deployment requirements

Under 802.3at PoE+ power, the AP can operate at full capability. Under 802.3af power, the AP may reduce radio capability to 2x2, reduce Ethernet operation to 1G, and disable USB.

Physical specifications:

```
Dimensions: 8.0 x 8.0 x 1.5 in.
Dimensions: 20.3 x 20.3 x 3.8 cm
Weight: 1.98 lb / 0.9 kg
```

Memory:

```
DRAM: 2048 MB
Flash: 1024 MB
```

Operating environment:

```
Temperature: 32°F to 122°F / 0°C to 50°C
Humidity: 10% to 90% non-condensing
```

Important performance note:

Above 40°C ambient temperature, the access point may reduce operation, including radio mode, Ethernet speed, and USB availability.

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4. Ports, Buttons, and Indicators

The Cisco Catalyst 9115AXI includes physical interfaces and indicators used for power, data, local management, reset, and status monitoring.

4.1 Multigigabit Ethernet Port

The main Ethernet port is an RJ-45 multigigabit uplink.

It is used for:

- Wired network uplink
- Data traffic
- PoE or PoE+ power input
- Controller communication
- Client traffic forwarding
- CAPWAP tunnel connectivity to the wireless controller

Supported Ethernet speeds include:

- 100 Mbps
- 1 Gbps
- 2.5 Gbps

Actual speed depends on:

- Switch port capability
- Cable quality
- Power mode
- Negotiation
- Configuration
- Environmental conditions

For best performance, connect the AP to a PoE+ switch port that supports multigigabit Ethernet, or use a supported Cisco injector.

4.2 RJ-45 Console Port

The RJ-45 console port is used for local management and troubleshooting.

Console access is useful when:

- The AP does not join the controller
- The AP is stuck during boot
- The AP has an IP addressing issue
- Remote management is unavailable
- Logs are needed during startup
- Reset or recovery must be verified

4.3 USB 2.0 Port

The USB 2.0 port can support IoT and application-related functions depending on software and deployment. However, the USB port may be disabled when the AP is powered by 802.3af PoE.

For full USB availability, use 802.3at PoE+.

4.4 Mode / Reset Button

The Mode/Reset button is used for reset and recovery operations.

It may be used when:

- Management access is lost
- The AP needs to return to factory defaults
- Configuration is corrupted
- Re-provisioning is required
- The AP must rejoin a controller from a clean state

The reset button should be used carefully because factory reset clears local configuration and requires the AP to be provisioned again.

4.5 Status LED

The status LED reports access point state.

It can indicate:

- Boot process
- Power status
- Controller join state
- Normal operation
- Warning state
- Error state
- Recovery or reset activity

During troubleshooting, always check the LED state before moving to deeper network or controller analysis.

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5. Installation and Setup Notes

Before installing the access point, confirm the model, power source, mounting bracket, switch port capability, cabling, and controller compatibility.

Basic installation steps:

1. Confirm that the device is Cisco Catalyst 9115AXI.
2. Confirm that the installation location is indoor.
3. Verify that the location is within the required temperature and humidity range.
4. Mount the AP on a ceiling or wall using the proper bracket.
5. Connect the Ethernet uplink to a PoE+ switch or supported Cisco injector.
6. Confirm that the switch port provides 802.3at PoE+ for full functionality.
7. Confirm that the switch port supports the required Ethernet speed.
8. Ensure that LLDP/CDP is enabled where required for proper power negotiation.
9. Confirm that the AP can reach the wireless controller.
10. Confirm that the controller supports the AP model and required software release.
11. Verify that the AP joins the controller.
12. Assign the AP to the correct site, policy, tag, profile, or WLAN configuration depending on controller design.
13. Confirm that the required SSIDs are broadcasting.
14. Test wireless client association and traffic.

The AP should be placed where wireless coverage is needed and should not be hidden inside metal cabinets, above dense obstructions, or near sources of heavy interference.

Recommended mounting locations:

- Ceiling center areas
- Open office ceilings
- Meeting room ceilings
- Hallways with planned coverage
- Indoor high-density spaces

Avoid installing near:

- Large metal surfaces
- Electrical panels
- Heavy machinery
- Microwave ovens
- Thick concrete obstructions
- Areas above temperature limits
- Outdoor or wet locations

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6. Power Requirements and PoE Behavior

The Catalyst 9115AXI requires correct PoE power for full performance.

6.1 Full Power Mode

For full operation, use 802.3at PoE+.

Full operation allows:

- 4x4 radio capability
- 2.5G Ethernet uplink capability
- USB availability
- Full AP feature operation under supported conditions

6.2 Reduced Power Mode

If the AP is powered by 802.3af PoE, it may enter reduced functionality mode.

Reduced mode may include:

- Radios reduced to 2x2
- Ethernet link reduced to 1G
- USB port disabled

This does not always mean the AP is faulty. It usually means the AP is not receiving enough power for full operation.

6.3 High Temperature Reduced Operation

If the ambient temperature is above 40°C, the AP may also reduce performance.

Possible reduced behavior:

- 2x2 radio operation
- 1G Ethernet link
- USB disabled

To restore full operation, verify the AP is powered by 802.3at PoE+ and that the ambient temperature is within the recommended full-performance range.

6.4 Power Source Options

The AP can be powered through:

- PoE+ capable Cisco switch
- Supported Cisco PoE injector
- Compatible power source approved for the deployment

Best practice:

Use a Cisco Catalyst PoE+ switch port when possible, because it simplifies cabling, power management, and monitoring.

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7. Switch Port Configuration for the Access Point

The AP connects to an access switch through an Ethernet port.

In many deployments, the AP switch port is configured as a trunk because the AP may carry multiple SSIDs mapped to different VLANs.

Example trunk port for AP:

```
configure terminal
interface GigabitEthernet1/0/10
 description Cisco Catalyst 9115AXI Access Point
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30,99
 power inline auto
 spanning-tree portfast trunk
 no shutdown
exit
```

Where VLANs may represent:

```
VLAN 10 - Corporate users
VLAN 20 - Guest Wi-Fi
VLAN 30 - IoT devices
VLAN 99 - Management / AP management
```

In simpler deployments, if the wireless controller handles tunneling and local VLANs are not required at the AP switch port, the AP may be placed in an access VLAN used for AP management.

Example access port:

```
configure terminal
interface GigabitEthernet1/0/10
 description Cisco Catalyst 9115AXI Access Point
 switchport mode access
 switchport access vlan 99
 power inline auto
 spanning-tree portfast
 no shutdown
exit
```

Verify switch port status:

```
show interfaces status
show power inline
```

```
show power inline GigabitEthernet1/0/10
show running-config interface GigabitEthernet1/0/10
```

Expected result:

- Port is connected
- PoE is delivering power
- AP receives an IP address
- AP reaches the controller
- AP joins successfully

Important note:

The correct switch port mode depends on wireless design, controller mode, VLAN mapping, and site policy.

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8. Controller Join and Wireless Setup

The Catalyst 9115AXI normally joins a Cisco Wireless LAN Controller.

The AP must be able to discover and reach the controller. Discovery may happen through network configuration such as DHCP, DNS, or previously stored controller information, depending on deployment.

Controller join requires:

- AP has power
- AP has an IP address
- AP can reach the controller
- AP model is supported by the controller
- Controller software version supports the AP
- Time, certificates, and regulatory domain are valid
- Network path between AP and controller is not blocked
- Required CAPWAP communication is allowed

Common controller-side checks:

```
show ap summary
show ap name <AP-NAME> config general
show wireless client summary
```

Useful AP-side console checks may include:

```
show version
show interfaces
show ip interface brief
show logging
```

Exact commands may differ depending on AP software mode and controller version.

After the AP joins the controller, the administrator should verify:

- AP appears in controller inventory
- AP has the correct name
- AP is assigned to the correct site or policy
- Radio interfaces are enabled
- SSIDs are mapped correctly
- Client VLANs are correct
- WLAN security settings are correct
- Clients can connect
- Clients receive valid IP addresses
- Clients can reach required network resources

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9. SSID and Wireless Service Operation

The AP broadcasts SSIDs configured through the wireless controller or Embedded Wireless Controller.

Common SSID types:

- Corporate SSID
- Guest SSID
- IoT SSID
- Voice SSID
- Meeting room SSID
- Management or testing SSID

Each SSID may have:

- Authentication method
- Encryption policy
- VLAN mapping
- Access policy
- QoS policy
- Client limit
- Band steering
- Radio policy
- Security filtering

Normal wireless service requires:

- AP joined to controller
- WLAN/SSID enabled
- Radio enabled
- Correct policy applied
- Correct VLAN available
- DHCP available for clients
- Authentication service reachable if used

- No major RF interference

If the AP is powered and joined but not broadcasting an SSID, check the controller configuration first.

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10. Normal Operation

During normal operation:

- AP receives PoE or PoE+ power
- Status LED shows normal operation
- Ethernet uplink is active
- AP receives an IP address
- AP joins the wireless controller
- Radios are operational
- Configured SSIDs are broadcasting
- Wireless clients can associate
- Clients receive valid IP addresses
- Clients can reach the network
- Controller can monitor AP health
- No repeated AP join/disconnect events appear in logs
- PoE power level matches expected operation

Under full PoE+ conditions:

- AP can operate with full 4x4 radio capability
- Ethernet uplink can operate at up to 2.5G where supported
- USB can remain enabled if supported by software and configuration

Useful verification checks:

```
Check AP LED status
Check switch port link
Check PoE status
Check AP IP address
Check AP controller join state
Check SSID broadcast
Check wireless client association
Check client DHCP address
Check client internet or LAN reachability
```

Useful switch-side commands:

```
show interfaces status
show power inline
show power inline <interface>
show running-config interface <interface>
show logging
```

Useful controller-side commands:

```
show ap summary
show wireless client summary
```

---

11. Troubleshooting

Troubleshooting should move from the physical layer upward:

1. Power source
2. PoE budget
3. Cabling
4. Switch port status
5. VLAN or trunk configuration
6. IP addressing
7. Controller reachability
8. Controller compatibility
9. AP join process
10. SSID configuration
11. Client authentication
12. RF/interference
13. Software or hardware fault

11.1 Access Point Does Not Power On

Symptom:

- No status LED
- AP does not boot
- Switch does not show PoE delivery
- No console output

Possible causes:

- Switch port does not provide PoE
- PoE is disabled on the switch port
- PoE budget is exhausted
- Cable is faulty

- Injector is not connected correctly
- Injector does not provide enough power
- AP hardware fault

Switch-side checks:

```
show power inline
show power inline <interface>
show interfaces status
show running-config interface <interface>
```

Fix actions:

```
configure terminal
interface <interface>
 power inline auto
 no shutdown
exit
```

Additional checks:

- Replace the Ethernet cable.
- Try another PoE+ switch port.
- Use an 802.3at PoE+ source.
- Verify the injector model if using an injector.
- Confirm the switch has enough available PoE budget.

Resolution:

Once proper PoE is available, the AP should power on and begin booting.

11.2 AP Powers On but Runs in Reduced Mode

Symptom:

- Radio capability appears reduced
- AP operates as 2x2 instead of 4x4
- Ethernet link is 1G instead of 2.5G
- USB port is disabled
- Controller reports power limitation

Possible causes:

- AP is powered by 802.3af instead of 802.3at PoE+
- PoE negotiation failed
- LLDP/CDP is disabled or not working
- Switch cannot provide enough power

- Ambient temperature is too high
- Power injector is insufficient

Checks:

```
show power inline
show power inline <interface>
show interfaces status
```

Fix actions:

- Use 802.3at PoE+.
- Enable proper power negotiation on the switch.
- Check LLDP/CDP.
- Use a supported Cisco injector.
- Reduce heat around the AP.
- Move the AP to an environment within temperature limits.
- Confirm the switch port supports required power.

Expected result:

The AP should return to full operation when sufficient power and acceptable temperature conditions are restored.

11.3 Ethernet Uplink Is Down

Symptom:

- AP has no network connectivity
- Switch port shows not connected
- AP does not receive IP address
- AP cannot join controller

Possible causes:

- Bad Ethernet cable
- Cable not fully seated
- Wrong switch port
- Switch port shutdown
- Port security or err-disable state
- Failed AP Ethernet port
- Failed switch port

Switch checks:

```
show interfaces status
show interfaces <interface>
```

```
show logging
show running-config interface <interface>
```

Fix:

```
configure terminal
interface <interface>
 no shutdown
exit
```

Additional fix actions:

- Replace cable.
- Move AP to another switch port.
- Check for err-disable.
- Verify speed negotiation.
- Check patch panel path.
- Test AP with a known-good cable and port.

11.4 AP Does Not Receive IP Address

Symptom:

- AP powers on
- Switch port is up
- AP does not get an IP address
- AP cannot discover controller

Possible causes:

- Wrong VLAN
- DHCP server unreachable
- DHCP scope exhausted
- Trunk does not allow AP management VLAN
- Access port is assigned to wrong VLAN
- DHCP relay missing
- Gateway down

Switch-side checks:

```
show vlan brief
show interfaces trunk
show mac address-table interface <interface>
show running-config interface <interface>
```

Controller or network checks:

```
Check DHCP scope
Check default gateway
Check AP management VLAN
Check DHCP relay/helper on router or Layer 3 interface
Check routing between AP subnet and controller
```

Fix actions:

- Assign correct access VLAN or trunk VLAN.
- Allow AP management VLAN on trunk.
- Confirm DHCP scope is available.
- Confirm gateway is reachable.
- Confirm DHCP relay is configured if DHCP server is remote.

Expected result:

The AP receives a valid IP address and begins controller discovery.

11.5 AP Does Not Join Controller

Symptom:

- AP powers on
- AP has network connectivity
- AP does not appear on controller
- AP repeatedly tries to join and fails
- SSIDs are not broadcast

Possible causes:

- Controller unreachable
- Wrong VLAN or routing
- DNS/DHCP discovery issue
- Controller software does not support AP model
- AP software compatibility issue
- CAPWAP blocked by firewall or ACL
- Certificate, time, or regulatory domain issue
- AP already joined to another controller
- Wrong site/controller configuration

Controller checks:

```
show ap summary
show ap join stats summary
show ap join stats detailed <AP-MAC>
```

Network checks:

```
Ping AP gateway
Ping controller from AP network
Check routing between AP and controller
Check firewall or ACL rules
Check DHCP option or DNS discovery if used
Check controller software version
```

Fix actions:

- Ensure AP and controller can communicate.
- Verify controller supports the AP model.
- Confirm required software version.
- Check controller discovery method.
- Check AP management VLAN.
- Remove network blocks between AP and controller.
- Reboot AP after correcting network/controller settings if needed.

Expected result:

The AP joins the controller and appears in the controller AP inventory.

11.6 AP Joined but SSID Is Not Broadcasting

Symptom:

- AP appears on controller
- AP is powered
- Clients cannot see SSID
- WLAN is missing from the air

Possible causes:

- WLAN/SSID disabled
- AP not assigned to correct policy/profile/tag
- Radio disabled
- SSID not mapped to this AP group/site
- AP in wrong location or policy
- Regulatory domain issue
- Controller configuration error

Controller checks:

```
show ap summary
show wlan summary
show ap name <AP-NAME> config general
```

Fix actions:

- Enable WLAN/SSID.
- Assign AP to correct policy or site.
- Verify radio status.
- Verify SSID-to-policy mapping.
- Confirm SSID is allowed on the AP.
- Check controller logs.

Expected result:

The AP broadcasts the configured SSID and clients can discover it.

11.7 Clients Cannot Connect to Wi-Fi

Symptom:

- SSID is visible
- Client fails to connect
- Authentication fails
- Client disconnects quickly

Possible causes:

- Wrong password
- Authentication server unreachable
- 802.1X/RADIUS issue
- WLAN security mismatch
- Client device incompatibility
- Client blocked by policy
- Maximum client limit reached
- Poor signal
- RF interference

Controller checks:

```
show wireless client summary
show wireless client mac-address <CLIENT-MAC> detail
```

Network checks:

```
Check WLAN security settings
Check RADIUS/authentication server
Check client VLAN
Check DHCP server
Check signal strength
Check RF interference
```

Fix actions:

- Confirm correct SSID password or credentials.
- Check RADIUS server reachability if using 802.1X.
- Confirm WLAN security policy.
- Check client compatibility.
- Move client closer to AP.
- Check RF interference.
- Review controller client logs.

11.8 Clients Connect but Have No Internet or LAN Access

Symptom:

- Client connects to SSID
- Client receives weak or no network access
- Client cannot reach gateway
- Client cannot browse internet
- Client cannot access internal systems

Possible causes:

- Client did not receive DHCP address
- Wrong VLAN mapping
- Gateway unreachable
- ACL blocks traffic
- Firewall policy blocks traffic
- Trunk missing client VLAN
- NAT or routing issue upstream
- DNS issue

Checks:

```
Check client IP address
Check default gateway
Check DNS server
Check VLAN mapping
Check switch trunk allowed VLANs
```

```
Check router/firewall routing
Check DHCP scope
```

Switch-side commands:

```
show interfaces trunk
show vlan brief
show mac address-table
```

Fix actions:

- Correct WLAN-to-VLAN mapping.
- Allow required VLANs on trunk.
- Check DHCP and gateway.
- Check router/firewall policies.
- Confirm DNS configuration.
- Test ping to gateway and external address.

11.9 Slow Wireless Performance

Symptom:

- Wi-Fi works but is slow
- High latency
- Video calls freeze
- File downloads are slow
- Clients disconnect intermittently

Possible causes:

- AP running in reduced power mode
- 802.3af power instead of PoE+
- Ethernet uplink negotiated at lower speed
- RF interference
- Too many clients on one AP
- Poor AP placement
- Channel congestion
- Low signal strength
- Wrong channel width
- High retransmissions
- Client device limitation

Checks:

```
Check AP power mode
Check Ethernet uplink speed
```

```
Check client count
Check signal strength
Check channel utilization
Check RF interference
Check controller performance reports
```

Switch checks:

```
show power inline <interface>
show interfaces status
show interfaces <interface>
```

Fix actions:

- Use PoE+.
- Ensure uplink negotiates correctly.
- Improve AP placement.
- Reduce interference.
- Balance clients across APs.
- Tune channel and power settings through controller.
- Add more APs if coverage or capacity is insufficient.

11.10 Intermittent AP Disconnects

Symptom:

- AP joins and disconnects repeatedly
- SSID disappears temporarily
- Clients disconnect randomly
- Controller logs repeated AP join events

Possible causes:

- Unstable PoE
- Bad cable
- Switch port errors
- DHCP lease issues
- Controller reachability problem
- Network instability
- Software bug
- AP overheating
- Power budget changes

Checks:

```
show interfaces <interface>
show interfaces counters errors
show power inline <interface>
show logging
```

Controller checks:

```
show ap summary
show ap join stats detailed <AP-MAC>
```

Fix actions:

- Replace cable.
- Move AP to another port.
- Verify PoE stability.
- Check switch logs.
- Check AP temperature/environment.
- Verify controller reachability.
- Upgrade to supported software if needed.

11.11 Console or Management Access Problem

Symptom:

- AP is powered
- AP may or may not join controller
- Admin cannot access AP locally or remotely
- Controller information is unavailable

Possible causes:

- Wrong console cable
- Wrong terminal settings
- AP not booting
- AP not reachable through network
- Controller access problem
- Management VLAN issue

Fix actions:

- Use the RJ-45 console port.
- Check terminal software and cable.
- Watch boot messages.
- Verify AP IP address.
- Verify controller status.
- Verify switch port and VLAN.

Useful console checks may include:

```
show version
show interfaces
show logging
```

Commands depend on AP software and operating mode.

11.12 USB Port Not Working

Symptom:

- USB device is not detected
- USB-dependent service unavailable
- Controller reports USB disabled

Possible causes:

- AP powered by 802.3af
- AP in reduced power mode
- High temperature reduced operation
- USB disabled by configuration
- Unsupported USB device
- Software limitation

Fix actions:

- Use 802.3at PoE+.
- Verify AP is not in reduced power mode.
- Check temperature.
- Verify controller configuration.
- Confirm USB use case is supported.

11.13 Software or Firmware Issue

Symptom:

- AP does not boot normally
- AP fails to join controller after upgrade
- Controller reports incompatible AP image
- AP reloads repeatedly

Possible causes:

- Unsupported controller version
- Unsupported AP image
- Corrupt image
- Failed upgrade

- Controller/AP software mismatch

Checks:

```
Check AP model support
Check controller software version
Check AP software version
Check release notes
Check controller logs
```

Fix actions:

- Use supported controller software.
- Upgrade controller to a compatible release.
- Allow AP to download the correct image.
- Follow Cisco recovery procedure if AP cannot boot normally.
- Use console logs to identify boot failure.

---

12. Factory Reset and Recovery

Factory reset returns the access point to a clean state and clears local configuration.

Factory reset may be used when:

- AP cannot be managed
- AP has incorrect stored controller information
- AP configuration is corrupted
- AP is being moved to another site
- AP must rejoin a different controller
- Troubleshooting requires a clean provisioning state

General reset method:

```
1. Disconnect power from the AP.
2. Press and hold the Mode/Reset button.
3. Reconnect power while holding the button.
4. Continue holding until the reset process starts according to the LED behavior.
5. Release the button.
6. Allow the AP to boot.
7. Re-provision the AP and allow it to join the controller.
```

Important warning:

Factory reset clears local AP configuration. After reset, the AP must obtain network connectivity and rejoin the controller before it can provide wireless service.

After reset, verify:

```
AP powers on
AP receives IP address
AP discovers controller
AP joins controller
Correct SSIDs are applied
Clients can connect
```

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13. Maintenance and Best Practices

Recommended practices:

- Use 802.3at PoE+ for full AP functionality.
- Use a PoE+ switch port whenever possible.
- Verify PoE budget before installing many APs.
- Use Cat5e or better cabling for multigigabit operation.
- Enable LLDP/CDP where required for power negotiation.
- Keep controller and AP software on a supported release.
- Mount the AP indoors only.
- Keep the AP within temperature and humidity limits.
- Avoid installing near metal obstacles or interference sources.
- Keep the AP clean and dust-free.
- Monitor AP status from the controller.
- Monitor client count and channel utilization.
- Document AP name, location, switch port, and MAC address.
- Label the switch port used by the AP.
- Avoid unnecessary power cycling.
- Use controller-based monitoring for performance and health.
- Plan firmware upgrades during maintenance windows.
- Keep deployment maps updated.

Useful maintenance checks:

```
Check AP online status
Check AP uptime
Check AP power mode
Check client count
Check SSID status
Check channel utilization
Check interference
Check switch port errors
Check PoE draw
Check controller logs
```

Useful switch-side commands:

```
show interfaces status
show power inline
show power inline <interface>
show interfaces <interface>
show interfaces counters errors
show logging
```

Useful controller-side checks:

```
show ap summary
show wireless client summary
```

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14. Quick Reference

Power Check

```
Confirm PoE+ source
Check switch PoE budget
Check AP status LED
Check power injector if used
Check switch port power status
```

Switch Port Check

```
show interfaces status
show power inline
show power inline <interface>
show running-config interface <interface>
```

Uplink Check

```
show interfaces <interface>
show interfaces counters errors
show logging
```

VLAN / Trunk Check

```
show vlan brief
show interfaces trunk
```

```
show running-config interface <interface>
```

AP Controller Join Check

```
show ap summary
show ap name <AP-NAME> config general
```

Wireless Client Check

```
show wireless client summary
show wireless client mac-address <CLIENT-MAC> detail
```

Basic AP Switch Port as Trunk

```
interface GigabitEthernet1/0/10
 description Cisco Catalyst 9115AXI Access Point
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30,99
 power inline auto
 spanning-tree portfast trunk
 no shutdown
```

Basic AP Switch Port as Access Port

```
interface GigabitEthernet1/0/10
 description Cisco Catalyst 9115AXI Access Point
 switchport mode access
 switchport access vlan 99
 power inline auto
 spanning-tree portfast
 no shutdown
```

Full Functionality Requirements

```
Use 802.3at PoE+
Use supported switch or injector
Use proper Ethernet cabling
Keep ambient temperature within limits
Verify controller software support
```

Reduced Functionality Indicators

```
2x2 radio operation
1G Ethernet link instead of 2.5G
USB disabled
Controller reports insufficient power
High temperature condition
```

Factory Reset / Recovery

```
Power off AP
Hold Mode/Reset button
Power on while holding button
Wait for reset indication
Release button
Allow AP to boot and rejoin controller
```

Restart

```
Power-cycle through PoE switch port
Or shut/no shut the switch port during maintenance
Avoid unnecessary restarts
```

Example switch port restart:

```
configure terminal
interface GigabitEthernet1/0/10
 shutdown
 no shutdown
exit
```